

Isospectral Arithmetic Fiber Retractions Across the $1/h < \Re s \leq 1$ Similarity Gap

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Abstract

For every integer $h \geq 2$, we compare two idempotent arithmetic retractions onto the h -free integers: one caps each prime exponent at $h - 1$, while the other reduces it modulo h . Weighting the induced basis maps on $\ell^2(\mathbb{N})$ by $n^{-s/2}$ produces compact operators $S_{h,s}$ and $M_{h,s}$ on their respective domains $\Re s > 0$ and $\Re s > 1/h$. Their simple nonzero eigenvalues coincide, and so do all common legal power traces and integer-order regularized determinants. Nevertheless, $S_{h,s}$ is boundedly similar to a compact normal operator exactly when $\Re s > 1$, whereas $M_{h,s}$ is so similar throughout its bounded domain. Thus the full band $1/h < \Re s \leq 1$ is isospectral but not similar.

The discrepancy is quantified in three ways. Saturated Riesz projections have an exact primordial optimizer and subcritical, critical, and supercritical maximal-order regimes. Both singular-value sequences satisfy Weyl laws with explicit Euler-product constants, meeting at the exact crossover $C_{h,1} = D_{h,1} = 1$. Finally, the self-commutators have the sharp wall $\Re(s)q = 1$, distinct from the operator Schatten wall $\Re(s)q = 2$; the case $h = 2$ requires a separate endpoint witness and admits an exact Hilbert–Schmidt Euler identity. Independent finite and analytic recomputations are reported only as implementation checks: all infinite claims are established by proof.

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1 Introduction

Compact operators can share every nonzero eigenvalue and still have incompatible geometry. The simplest obstruction is already visible in a rank-one block: the eigenvalue records the action on its eigenline, whereas the angle between the range and kernel controls the norm of its Riesz projection. The point of this paper is that two elementary arithmetic retractions organize that obstruction into an exact family of phase transitions.

Fix $h \geq 2$. Let $\mathcal{F}_h$ be the positive integers whose prime exponents are strictly smaller than h , and define

$$\tau_h(n) = \prod_p p^{\min\{v_p(n), h-1\}}, \quad \omega_h(n) = \prod_p p^{v_p(n) \bmod h}. \quad (1)$$

Both maps retract $\mathbb{N}$ onto $\mathcal{F}_h$. On the finitely supported sequences set

$$S_{h,s}e_n = n^{-s/2}e_{\tau_h(n)}, \quad M_{h,s}e_n = n^{-s/2}e_{\omega_h(n)}, \quad \sigma = \Re s. \quad (2)$$

Here and below $n^{-s/2} = \exp(-(s/2) \log n)$ with the real logarithm. The formulas in equation (2) are algebraic until boundedness is proved. The symbols $S_{h,s}$ and $M_{h,s}$ denote operators on $\ell^2(\mathbb{N})$ only on their respective bounded domains; every spectral, ideal, trace, determinant, similarity, and commutator assertion is confined to those domains. For $0 < q < 1$, $\mathcal{S}_q$ denotes the usual quasi-Schatten ideal.

The common fixed points in $\mathcal{F}_h$ force common eigenlines. The fibers above those points are very different: saturation can increase only primes already at exponent $h-1$, whereas reduction modulo h permits an arbitrary h th-power multiplier. That distinction changes fiber mass and block angle without changing the eigenvalue on the fixed point.

Theorem 1.1 (Paired arithmetic retraction theorem). *Let $h \geq 2$, $s \in \mathbb{C}$, and $\sigma = \Re s$.*

- (i) *$S_{h,s}$ extends boundedly exactly for $\sigma > 0$, and $M_{h,s}$ extends boundedly exactly for $\sigma > 1/h$. Each extension is compact throughout its bounded domain.*
- (ii) *For $k \geq 1$ and $0 < q < \infty$,*

$$\begin{aligned} S_{h,s}^k &\in \mathcal{S}_q \iff k\sigma q > 2, \\ M_{h,s}^k &\in \mathcal{S}_q \iff \sigma > 1/h \text{ and } k\sigma q > 2. \end{aligned}$$

- (iii) *On each bounded domain the simple nonzero eigenvalues are $\lambda_m = m^{-s/2}$, $m \in \mathcal{F}_h$. On the common domain $\sigma > 1/h$, if $k\sigma > 2$, then*

$$\mathrm{Tr}(S_{h,s}^k) = \mathrm{Tr}(M_{h,s}^k) = \frac{\zeta(ks/2)}{\zeta(hks/2)}. \quad (3)$$

If $r \geq 1$ is an integer, $\sigma > 1/h$, and $r\sigma > 2$, then

$$\det_r(I - zS_{h,s}) = \det_r(I - zM_{h,s}) \quad (4)$$

as entire functions of z . For $r = 1$, this is the ordinary Fredholm determinant and requires $\sigma > 2$.

(iv) *The exact normal-similarity domains are*

$$\begin{aligned} S_{h,s} \sim_{\text{bd}} \text{ a compact normal operator} &\iff \sigma > 1, \\ M_{h,s} \sim_{\text{bd}} \text{ a compact normal operator} &\iff \sigma > 1/h. \end{aligned}$$

Consequently $1/h < \sigma \leq 1$ is an isospectral, equal-determinant band in which $M_{h,s}$ is similar to normal and $S_{h,s}$ is not.

(v) *Let $P_k = p_1 \cdots p_k$, with $P_0 = 1$, and take the largest $k = k(x)$ for which $P_k^{h-1} \leq x$. The largest saturated Riesz norm among $m \in \mathcal{F}_h$, $m \leq x$, is attained at P_k^{h-1} . As $x \rightarrow \infty$ this maximum tends to $\sqrt{\zeta(\sigma)}$ if $\sigma > 1$; it is asymptotic to $\sqrt{e^\gamma \log \log x}$ if $\sigma = 1$; and if $0 < \sigma < 1$, its logarithm is asymptotic to*

$$\frac{(h-1)^{\sigma-1} (\log x)^{1-\sigma}}{2(1-\sigma) \log \log x}. \quad (5)$$

(vi) *The singular values, counted with multiplicity in decreasing order, satisfy*

$$\begin{aligned} s_n(S_{h,s}) &\sim \left(\frac{C_{h,\sigma}}{n} \right)^{\sigma/2} & (\sigma > 0), \\ s_n(M_{h,s}) &\sim \left(\frac{D_{h,\sigma}}{n} \right)^{\sigma/2} & (\sigma > 1/h), \\ |\lambda_n| &\sim \left(\frac{1/\zeta(h)}{n} \right)^{\sigma/2}, \end{aligned}$$

where

$$C_{h,\sigma} = \prod_p (1 - p^{-1}) \left[\sum_{e=0}^{h-2} p^{-e} + p^{-(h-1)} (1 - p^{-\sigma})^{-1/\sigma} \right], \quad (6)$$

$$D_{h,\sigma} = \frac{\zeta(h\sigma)^{1/\sigma}}{\zeta(h)}. \quad (7)$$

The constants meet at $C_{h,\sigma} = D_{h,\sigma} = 1$ when $\sigma = 1$. No ordering between them is asserted away from that point.

(vii) *For $0 < q < \infty$,*

$$\begin{aligned} [S_{h,s}^*, S_{h,s}] &\in \mathcal{S}_q \iff \sigma q > 1, \\ [M_{h,s}^*, M_{h,s}] &\in \mathcal{S}_q \iff \sigma > 1/h \text{ and } \sigma q > 1. \end{aligned}$$

The theorem separates three ledgers. The cyclic ledger—eigenvalues, legal traces, and legal determinants—is common. The metric ledger—block singular values and their Weyl constants—is not. The angular ledger—Riesz projection norms and self-commutators—produces a further pair of sharp walls. In particular, equality of the regularized determinants in equation (4) is a negative control: it demonstrates how much geometry a complete nonzero-eigenvalue product can fail to see.

Relation to prior work. Weighted composition operators on weighted sequence spaces have general boundedness, compactness, closed-range, and essential-norm theories [LK15, AM23]. Carlson’s discrete setting gives spectral and commutant context for weighted composition operators on L^2 spaces [Car90]. Our basis maps are naturally viewed on the adjoint side of that general framework; this convention and the generic fiber-norm mechanism are not contributions. Power-free parts also have a classical arithmetic life independent of the operators considered here [dWvdW99]. The residual result is the paired all- h classification and its exact

arithmetic distortion laws, not a generic weighted-composition theorem and not a claim that the formulas detect the additive semantics of rational primes.

We use standard trace-ideal and regularized-determinant theory from [Sim05], classical prime asymptotics from [MV06], and Wiener–Ikehara in the form discussed in [Kor04]. The hypotheses needed here are checked explicitly.

Proof organization. Section 2 derives both fibers and their rank-one blocks. Section 3 proves existence, compactness, ideal membership, and the common cyclic ledger. Section 4 identifies the precise similarity gap; sections 5 to 7 quantify its maximal, asymptotic, and commutator manifestations. Independent recomputation is summarized only after the proofs in section 8. It checks the implementation but is not used to prove an endpoint or asymptotic statement.

2 Arithmetic fibers and rank-one blocks

Write

$$\mathcal{F}_h = \{m \in \mathbb{N} : v_p(m) < h \text{ for every prime } p\}, \quad J_h(m) = \{p : v_p(m) = h - 1\}.$$

The set $J_h(m)$ records the prime exponents at which saturation has lost information. It need not equal the set of all prime divisors of m .

Lemma 2.1 (Exact fibers). *For every $m \in \mathcal{F}_h$,*

$$\tau_h^{-1}(m) = \left\{ m \prod_{p \in J_h(m)} p^{r_p} : r_p \in \mathbb{N} \cup \{0\} \right\}, \quad (8)$$

$$\omega_h^{-1}(m) = \{ma^h : a \in \mathbb{N}\}. \quad (9)$$

Both maps fix m .

Proof. If $v_p(m) \leq h - 2$, the equality $\min\{v_p(n), h - 1\} = v_p(m)$ forces $v_p(n) = v_p(m)$. If $v_p(m) = h - 1$, it forces only $v_p(n) \geq h - 1$, leaving a nonnegative extra exponent at that same prime. No new prime can occur, which proves equation (8). On the other hand, $v_p(n) \equiv v_p(m) \pmod{h}$, with $0 \leq v_p(m) < h$, is equivalent to $v_p(n) = v_p(m) + hv_p(a)$ for a unique positive integer a . This proves equation (9). Taking every extra exponent to be zero shows that both maps fix m . $\square$

For $f = \tau_h$ or ω_h , let

$$\mathcal{H}_m^f = \overline{\text{span}}\{e_n : f(n) = m\}.$$

The fibers partition $\mathbb{N}$, so

$$\ell^2(\mathbb{N}) = \bigoplus_{m \in \mathcal{F}_h} \mathcal{H}_m^f. \quad (10)$$

Lemma 2.2 (Block ledger). *Let T denote either algebraic map in equation (2), and write $T_m = T|_{\mathcal{H}_m^f}$. Whenever the coefficient sequence of this block is square summable, T_m is rank one, its range is $\mathbb{C}e_m$, and its unique nonzero singular value $\rho_T(m)$ satisfies*

$$\rho_S(m)^2 = m^{-\sigma} \prod_{p \in J_h(m)} (1 - p^{-\sigma})^{-1}, \quad \sigma > 0, \quad (11)$$

$$\rho_M(m)^2 = m^{-\sigma} \zeta(h\sigma), \quad \sigma > 1/h. \quad (12)$$

Moreover, with $\lambda_m = m^{-s/2}$,

$$T_m e_m = \lambda_m e_m, \quad T_m^k = \lambda_m^{k-1} T_m \quad (k \geq 1). \quad (13)$$

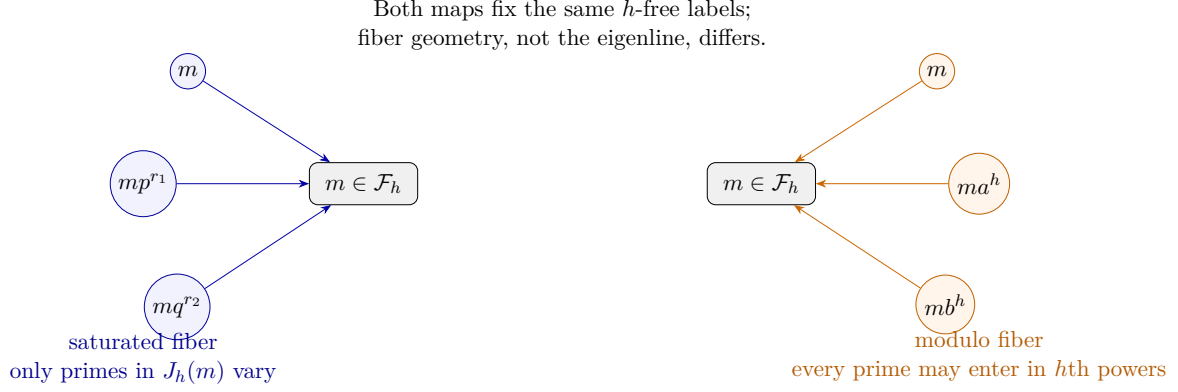


Figure 1: The common fixed label produces the same eigenline. The saturated fiber varies only primes in $J_h(m)$, whereas the modulo fiber permits an arbitrary h th-power multiplier.

Proof. For a finitely supported $x = \sum_n x_n e_n$ in the block,

$$T_m x = \left(\sum_{f(n)=m} x_n n^{-s/2} \right) e_m.$$

Thus the squared norm of the coefficient functional is $\sum_{f(n)=m} n^{-\sigma}$, and this is the only nonzero squared singular value. Applying lemma 2.1,

$$\begin{aligned} \sum_{\tau_h(n)=m} n^{-\sigma} &= m^{-\sigma} \prod_{p \in J_h(m)} \sum_{r \geq 0} p^{-r\sigma}, \\ \sum_{\omega_h(n)=m} n^{-\sigma} &= m^{-\sigma} \sum_{a \geq 1} a^{-h\sigma}, \end{aligned}$$

which gives equations (11) and (12) with their exact convergence conditions. Because m lies in its own fiber, $T_m e_m = \lambda_m e_m$. A rank-one map with range $\mathbb{C}e_m$ then obeys $T_m^2 = \lambda_m T_m$, and induction proves equation (13). $\square$

Two type distinctions will be used repeatedly. The eigenvalue λ_m depends on the full complex parameter s ; the block singular value depends only on σ . Also, the block index m is an h -free integer, not a prime. Confusing either pair erases the phenomenon we are trying to measure.

3 Existence, ideal membership, and the common cyclic ledger

The block decomposition converts operator questions into positive sums. This also makes every strict endpoint visible without analytic continuation or finite truncation.

Proposition 3.1 (Exact boundedness and compactness domains). *$S_{h,s}$ is bounded exactly when $\sigma > 0$, and $M_{h,s}$ is bounded exactly when $\sigma > 1/h$. Each bounded operator is compact.*

Proof. If $\sigma \leq 0$, the saturated block $m = p^{h-1}$ contains p^{h-1+r} for every $r \geq 0$, and its coefficient square mass diverges. Thus $S_{h,s}$ cannot be bounded. Suppose $\sigma > 0$. A nonsaturated local contribution to equation (11) is $p^{-e\sigma} < 1$. A saturated contribution is

$$b_p = p^{-(h-1)\sigma} (1 - p^{-\sigma})^{-1}.$$

Since $b_p \rightarrow 0$, only finitely many b_p exceed one; their product is a uniform bound for all squared block norms. Hence $S_{h,s}$ is bounded.

As $m \rightarrow \infty$ through $\mathcal{F}_h$, at least one prime divisor of m tends to infinity because every exponent is at most $h-1$. Its local contribution tends to zero, while all other contributions are bounded by the fixed product of the finitely many exceptional factors. Therefore $\rho_S(m) \rightarrow 0$. An orthogonal direct sum of rank-one blocks is compact exactly when its block norms tend to zero.

For $M_{h,s}$, the block at $m=1$ has squared mass $\zeta(h\sigma)$, which is finite exactly for $\sigma > 1/h$. On this domain $\rho_M(m) = \zeta(h\sigma)^{1/2} m^{-\sigma/2} \rightarrow 0$, giving both boundedness and compactness. $\square$

Proposition 3.2 (Powers and Schatten walls). *For $k \geq 1$ and $0 < q < \infty$,*

$$S_{h,s}^k \in \mathcal{S}_q \iff k\sigma q > 2,$$

$$M_{h,s}^k \in \mathcal{S}_q \iff \sigma > 1/h \text{ and } k\sigma q > 2.$$

Every equality endpoint fails.

Proof. For $k=1$, orthogonality and equation (11) give

$$\sum_{m \in \mathcal{F}_h} \rho_S(m)^q = \prod_p \left[1 + \sum_{e=1}^{h-2} p^{-e\sigma q/2} + p^{-(h-1)\sigma q/2} (1 - p^{-\sigma})^{-q/2} \right]. \quad (14)$$

When $h \geq 3$, the first nonconstant term is $p^{-\sigma q/2}$; when $h=2$, the saturated term is asymptotic to the same quantity. The positive Euler product converges exactly for $\sigma q/2 > 1$.

For M ,

$$\sum_{m \in \mathcal{F}_h} \rho_M(m)^q = \zeta(h\sigma)^{q/2} \sum_{m \in \mathcal{F}_h} m^{-\sigma q/2} = \zeta(h\sigma)^{q/2} \frac{\zeta(\sigma q/2)}{\zeta(h\sigma q/2)}. \quad (15)$$

The first factor imposes the boundedness condition $\sigma > 1/h$; the positive h -free sum converges exactly for $\sigma q/2 > 1$.

By equation (13), the unique singular value of T_m^k is $|\lambda_m|^{k-1} \rho_T(m)$. Repeating the two products replaces the leading exponent $\sigma q/2$ by $k\sigma q/2$ while leaving the modulo fiber factor $\zeta(h\sigma)^{q/2}$ unchanged. This proves the stated conditions. At equality, the leading positive prime sum is harmonic over the primes, so cancellation cannot rescue the endpoint. $\square$

Proposition 3.3 (Common spectrum and traces). *On either bounded domain, the nonzero spectrum is the simple set $\{m^{-s/2} : m \in \mathcal{F}_h\}$. On the common bounded domain, if $k\sigma > 2$, then equation (3) holds. For every integer $r \geq 1$ satisfying $\sigma > 1/h$ and $r\sigma > 2$, the regularized determinant identity equation (4) holds.*

Proof. Each block has the sole nonzero eigenvalue λ_m by lemma 2.2. Since $\sigma > 0$ on every bounded domain, $|\lambda_m| = m^{-\sigma/2}$ is strictly decreasing with m ; hence no two blocks contribute the same eigenvalue. Compactness adds only zero to the spectrum.

If $k\sigma > 2$, proposition 3.2 makes the relevant powers trace class, and absolute convergence permits the block traces to be summed:

$$\begin{aligned} \text{Tr}(T^k) &= \sum_{m \in \mathcal{F}_h} \lambda_m^k = \prod_p \sum_{e=0}^{h-1} p^{-eks/2} \\ &= \prod_p \frac{1 - p^{-hks/2}}{1 - p^{-ks/2}} = \frac{\zeta(ks/2)}{\zeta(hks/2)}. \end{aligned}$$

Table 1: The common cyclic ledger and the distinct metric/angular ledgers.

Quantity	Saturated $S_{h,s}$	Modulo $M_{h,s}$
Bounded domain	$\sigma > 0$	$\sigma > 1/h$
Nonzero eigenvalues	$m^{-s/2}, m \in \mathcal{F}_h$, simple	
Block singular square	$m^{-\sigma} \prod_{p \in J_h(m)} (1 - p^{-\sigma})^{-1}$	$m^{-\sigma} \zeta(h\sigma)$
Riesz norm	$\prod_{p \in J_h(m)} (1 - p^{-\sigma})^{-1/2}$	$\zeta(h\sigma)^{1/2}$
Similarity to normal	$\sigma > 1$	$\sigma > 1/h$

At $k\sigma = 2$, the modulus sum is the divergent h -free harmonic series, so the equality line is excluded.

The determinant statement follows from the common simple nonzero eigenvalues and standard regularized determinant theory [Sim05]; the exact product and its legality are recorded in appendix B. Nothing in that argument implies similarity. $\square$

4 Isospectrality without bounded similarity

The spectral idempotent for the nonzero eigenvalue in block m is

$$\Pi_{T,m} = \lambda_m^{-1} T_m. \quad (16)$$

It is an oblique projection onto $\mathbb{C}e_m$, and therefore

$$\|\Pi_{T,m}\| = \frac{\rho_T(m)}{|\lambda_m|}. \quad (17)$$

Combining this ratio with the two block masses yields

$$\|\Pi_{S,m}\| = \prod_{p \in J_h(m)} (1 - p^{-\sigma})^{-1/2}, \quad (18)$$

$$\|\Pi_{M,m}\| = \sqrt{\zeta(h\sigma)}. \quad (19)$$

The following block lemma supplies the converse that a mere appeal to necessary projection bounds would miss.

Lemma 4.1 (Uniform block diagonalization). *Let $T = \bigoplus_m T_m$ be a compact orthogonal direct sum of rank-one blocks. Suppose each block has nonzero eigenvalue λ_m with range $\mathbb{C}e_m$. Then T is boundedly similar to the normal diagonal operator with entries λ_m and zeros on the block kernels if and only if $\sup_m \|\lambda_m^{-1} T_m\| < \infty$.*

Proof. If $T = XNX^{-1}$ and N is normal, the nonzero spectral projections of N are orthogonal. Hence those of T have norms at most $\|X\|\|X^{-1}\|$.

Conversely, decompose a block as $\mathcal{H}_m = \mathbb{C}e_m \oplus K_m$. It has the matrix form

$$T_m = \begin{pmatrix} \lambda_m & \varphi_m \\ 0 & 0 \end{pmatrix}.$$

Set

$$Y_m = \begin{pmatrix} 1 & -\lambda_m^{-1} \varphi_m \\ 0 & I_{K_m} \end{pmatrix}.$$

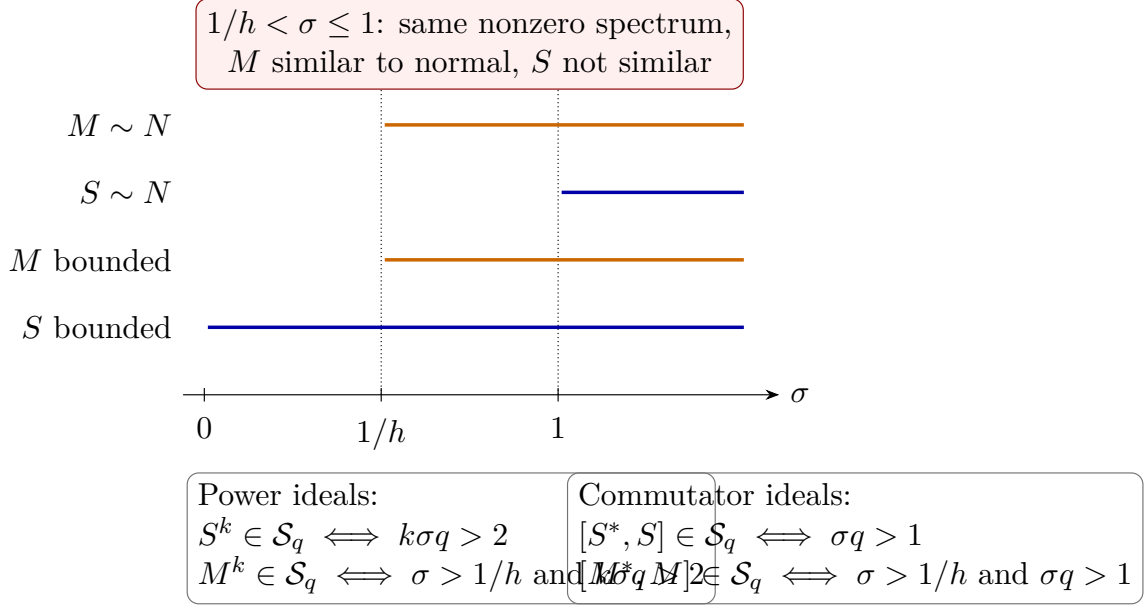


Figure 2: Exact existence, similarity, power-ideal, and commutator-ideal conditions. The horizontal bands encode only the fixed ordering $0 < 1/h \leq 1$; the ideal thresholds are stated algebraically because their relative positions depend on h, k, q . Every displayed wall is strict.

A direct multiplication gives $Y_m^{-1}T_mY_m = \text{diag}(\lambda_m, 0)$. The norm of $\lambda_m^{-1}\varphi_m$ is controlled by the norm of the idempotent $\lambda_m^{-1}T_m$; therefore the hypothesis uniformly bounds both Y_m and Y_m^{-1} . The direct sum $Y = \bigoplus_m Y_m$ is bounded and invertible and conjugates T to the stated diagonal. Since $|\lambda_m| \rightarrow 0$, that diagonal is compact and normal. A norm identity and explicit bounds are given in appendix A. $\square$

Theorem 4.2 (Exact similarity gap).

$$\begin{aligned}
 S_{h,s} \sim_{\text{bd}} \text{compact normal} &\iff \sigma > 1, \\
 M_{h,s} \sim_{\text{bd}} \text{compact normal} &\iff \sigma > 1/h.
 \end{aligned}$$

Proof. The modulo norm in equation (19) is a finite constant, independent of m , throughout the bounded domain. The lemma applies.

Every finite set of primes can occur as $J_h(m)$, by taking $m = \prod_{p \in E} p^{h-1}$. Consequently

$$\sup_{m \in \mathcal{F}_h} \|\Pi_{S,m}\| = \left(\prod_p (1 - p^{-\sigma})^{-1} \right)^{1/2}$$

as a finite value or $+\infty$. This is finite exactly for $\sigma > 1$. The lemma again gives both directions. $\square$

In the full interval $1/h < \sigma \leq 1$, both operators exist and have the same simple nonzero spectrum. Their common power traces and determinants also agree whenever the corresponding ideal condition is met. Yet one operator has uniformly controlled eigenprojections and the other does not. This is an obstruction to bounded similarity, not merely to unitary equivalence.

5 Primorial growth of saturated spectral projections

The failure of uniform similarity for $0 < \sigma \leq 1$ has a precise maximal order. Let $p_1 < p_2 < \dots$ be the primes and $P_k = \prod_{j \leq k} p_j$, with $P_0 = 1$. For $x \geq 1$, let $k(x)$ be the largest integer for

which

$$P_{k(x)}^{h-1} \leq x, \quad m_x = P_{k(x)}^{h-1}. \quad (20)$$

Proposition 5.1 (Exact optimizer). *For every $x \geq 1$ and $\sigma > 0$,*

$$\max_{\substack{m \leq x \\ m \in \mathcal{F}_h}} \|\Pi_{S,m}\| = \prod_{p|P_{k(x)}} (1 - p^{-\sigma})^{-1/2}, \quad (21)$$

and m_x is a maximizing label.

Proof. Suppose $m \leq x$ has $r = |J_h(m)|$ saturated primes. Their mandatory contribution to m gives

$$\prod_{p \in J_h(m)} p^{h-1} \leq m \leq x.$$

If $r \geq k(x) + 1$, the left side is at least $P_{k(x)+1}^{h-1} > x$, a contradiction. For fixed r , replacing a saturated prime by a smaller omitted prime decreases the arithmetic cost and increases its factor $(1 - p^{-\sigma})^{-1/2}$. Thus the best r -prime set is $\{p_1, \dots, p_r\}$. Adding another admissible saturated prime strictly increases the projection norm. The optimum therefore uses $r = k(x)$ and is attained at m_x . $\square$

The exact optimizer turns the similarity wall into a classical prime-product problem. Write $y = p_{k(x)}$. The prime number theorem gives

$$(h-1)\vartheta(y) \sim \log x, \quad y \sim \frac{\log x}{h-1}. \quad (22)$$

Theorem 5.2 (Three maximal-order regimes). *As $x \rightarrow \infty$,*

(a) *if $\sigma > 1$, then*

$$\max_{m \leq x, m \in \mathcal{F}_h} \|\Pi_{S,m}\| \longrightarrow \sqrt{\zeta(\sigma)};$$

(b) *if $\sigma = 1$, then*

$$\max_{m \leq x, m \in \mathcal{F}_h} \|\Pi_{S,m}\| \sim \sqrt{e^\gamma \log \log x};$$

(c) *if $0 < \sigma < 1$, then*

$$\log \max_{m \leq x, m \in \mathcal{F}_h} \|\Pi_{S,m}\| \sim \frac{(h-1)^{\sigma-1} (\log x)^{1-\sigma}}{2(1-\sigma) \log \log x}.$$

Proof. For $\sigma > 1$, the partial Euler products in equation (21) converge to $\prod_p (1 - p^{-\sigma})^{-1/2} = \sqrt{\zeta(\sigma)}$. At $\sigma = 1$, Mertens' product theorem gives

$$\prod_{p \leq y} (1 - p^{-1})^{-1/2} \sim \sqrt{e^\gamma \log y},$$

and equation (22) implies $\log y \sim \log \log x$.

For $0 < \sigma < 1$, expanding the logarithm and using the standard prime-sum asymptotic [MV06],

$$\begin{aligned} \log \prod_{p \leq y} (1 - p^{-\sigma})^{-1/2} &= \frac{1}{2} \sum_{p \leq y} -\log(1 - p^{-\sigma}) \\ &\sim \frac{1}{2} \sum_{p \leq y} p^{-\sigma} \sim \frac{y^{1-\sigma}}{2(1-\sigma) \log y}. \end{aligned}$$

Substituting equation (22) produces the factor $(h-1)^{\sigma-1}$. In particular, neither its reciprocal nor its deletion is compatible with the exact optimizer. $\square$

The proof uses no finite maximization. The finite optimizer rows reported in appendix D are implementation checks of proposition 5.1; they are not evidence for the asymptotic regimes.

6 Singular-value Weyl laws and the crossover

The saturated block mass is not a constant multiple of $m^{-\sigma}$, so ordinary h -free counting is insufficient. Define the positive generalized weight

$$w_{h,\sigma}(m) = m \prod_{p \in J_h(m)} (1 - p^{-\sigma})^{1/\sigma}. \quad (23)$$

Then equation (11) is exactly $\rho_S(m) = w_{h,\sigma}(m)^{-\sigma/2}$.

6.1 The saturated count

Consider the generalized Dirichlet series

$$F_{h,\sigma}(z) = \sum_{m \in \mathcal{F}_h} w_{h,\sigma}(m)^{-z} = \prod_p L_p(z), \quad (24)$$

where initially $\Re z > 1$ and

$$L_p(z) = \sum_{e=0}^{h-2} p^{-ez} + p^{-(h-1)z} (1 - p^{-\sigma})^{-z/\sigma}. \quad (25)$$

Factor the single zeta pole:

$$F_{h,\sigma}(z) = \zeta(z) G_{h,\sigma}(z), \quad G_{h,\sigma}(z) = \prod_p (1 - p^{-z}) L_p(z). \quad (26)$$

Lemma 6.1 (Local cancellation and strip). *$G_{h,\sigma}$ is holomorphic in*

$$\Re z > \theta_{h,\sigma} := \max \left\{ \frac{1}{h}, \frac{1 - \sigma}{h - 1} \right\} < 1. \quad (27)$$

At $z = 1$ it has the positive value $C_{h,\sigma}$ in equation (6).

Proof. On a compact subset of a right half-plane, expand the final term of equation (25) uniformly for large p . Multiplication by $1 - p^{-z}$ cancels the first-order p^{-z} term, leaving

$$(1 - p^{-z}) L_p(z) = 1 + O(p^{-h\Re z}) + O(p^{-(h-1)\Re z - \sigma}). \quad (28)$$

Both error sums converge locally uniformly precisely when $h\Re z > 1$ and $(h-1)\Re z + \sigma > 1$. The Weierstrass theorem for Euler products gives holomorphy in equation (27). Because $h \geq 2$ and $\sigma > 0$, the boundary lies strictly to the left of one. Substituting $z = 1$ gives equation (6); every local factor is positive. $\square$

There are at most finitely many local generalized weights below one. Indeed, the nonsaturated weights are $p^e > 1$, while the saturated local weight $p^{h-1}(1 - p^{-\sigma})^{1/\sigma}$ tends to infinity. Since exponents are bounded, the associated positive counting measure is locally finite.

Proposition 6.2 (Saturated Weyl law). *For $\sigma > 0$, let*

$$A_S(x) = \#\{m \in \mathcal{F}_h : w_{h,\sigma}(m) \leq x\}.$$

Then

$$A_S(x) \sim C_{h,\sigma} x, \quad s_n(S_{h,s}) \sim \left(\frac{C_{h,\sigma}}{n} \right)^{\sigma/2}. \quad (29)$$

Proof. By equation (26) and lemma 6.1, the series has a single pole at $z = 1$ with positive residue $G_{h,\sigma}(1) = C_{h,\sigma}$, while the remainder extends across the line $\Re z = 1$. Wiener–Ikehara for the nondecreasing positive counting measure therefore gives $A_S(x) \sim C_{h,\sigma}x$; see [Kor04]. The number of singular values at least t equals $A_S(t^{-2/\sigma})$. Monotone asymptotic inversion gives the second relation. The boundary regularity and the harmless finitely many weights below one are detailed in appendix C. $\square$

6.2 Modulo and eigenvalue counts

The elementary h -free density is

$$\#\{m \leq x : m \in \mathcal{F}_h\} \sim \frac{x}{\zeta(h)}. \quad (30)$$

For completeness, the identity $\mathbf{1}_{\mathcal{F}_h}(n) = \sum_{d^h|n} \mu(d)$ gives

$$\sum_{n \leq x} \mathbf{1}_{\mathcal{F}_h}(n) = \sum_{d \leq x^{1/h}} \mu(d) \left\lfloor \frac{x}{d^h} \right\rfloor = \frac{x}{\zeta(h)} + O_h(x^{1/h}).$$

Proposition 6.3 (Modulo and eigenvalue Weyl laws). *For $\sigma > 1/h$,*

$$s_n(M_{h,s}) \sim \left(\frac{D_{h,\sigma}}{n} \right)^{\sigma/2}, \quad D_{h,\sigma} = \frac{\zeta(h\sigma)^{1/\sigma}}{\zeta(h)}, \quad (31)$$

$$|\lambda_n| \sim \left(\frac{1/\zeta(h)}{n} \right)^{\sigma/2}. \quad (32)$$

Proof. By equation (12), $\rho_M(m) \geq t$ exactly when $m \leq \zeta(h\sigma)^{1/\sigma} t^{-2/\sigma}$. Applying equation (30) yields the singular-value count $D_{h,\sigma} t^{-2/\sigma}$. The same count without the fiber multiplier gives the eigenvalue constant $1/\zeta(h)$. Monotone inversion proves equations (31) and (32). $\square$

Corollary 6.4 (Exact crossover). *For every $h \geq 2$, $C_{h,1} = D_{h,1} = 1$.*

Proof. At $\sigma = 1$, the bracket in equation (6) is

$$\sum_{e=0}^{h-2} p^{-e} + p^{-(h-1)}(1 - p^{-1})^{-1} = \sum_{e=0}^{\infty} p^{-e} = (1 - p^{-1})^{-1}.$$

Every Euler factor in $C_{h,1}$ is therefore one. Independently, $D_{h,1} = \zeta(h)/\zeta(h) = 1$. $\square$

The crossover is an equality, not a sign-change theorem: we make no claim about the ordering of $C_{h,\sigma}$ and $D_{h,\sigma}$ for $\sigma \neq 1$. None of the asymptotic or endpoint statements in this section depends on finite records.

7 Self-commutator ideals

The operator ideal wall arises from the first power of a block singular value. A self-commutator sees its square and hence produces a different threshold.

Lemma 7.1 (Rank-one self-commutator). *Let $T = \rho u \otimes v$, where u, v are unit vectors, and let a be the modulus of the unique nonzero eigenvalue. The two possibly zero singular values of $T^*T - TT^*$ are equal to*

$$c = \rho^2 \sqrt{1 - a^2/\rho^2}. \quad (33)$$

Proof. $T^*T = \rho^2 v \otimes v$ and $TT^* = \rho^2 u \otimes u$. Their difference is self-adjoint and traceless on $\text{span}\{u, v\}$. Its determinant there is $-\rho^4(1 - |\langle u, v \rangle|^2)$. Since $a = \rho|\langle u, v \rangle|$, the two eigenvalues are $\pm c$, proving the claim. $\square$

For the arithmetic blocks, equations (11) and (12) gives the exact angle ratios

$$\frac{|\lambda_m|^2}{\rho_S(m)^2} = \prod_{p \in J_h(m)} (1 - p^{-\sigma}), \quad (34)$$

$$\frac{|\lambda_m|^2}{\rho_M(m)^2} = \frac{1}{\zeta(h\sigma)}. \quad (35)$$

Theorem 7.2 (Exact commutator wall). *For $0 < q < \infty$,*

$$\begin{aligned} [S_{h,s}^*, S_{h,s}] &\in \mathcal{S}_q \iff \sigma q > 1, \\ [M_{h,s}^*, M_{h,s}] &\in \mathcal{S}_q \iff \sigma > 1/h \text{ and } \sigma q > 1. \end{aligned}$$

Proof. Let c_m be the quantity in equation (33) for block m . For S , $c_m \leq \rho_S(m)^2$, so

$$\sum_m c_m^q \leq \sum_m \rho_S(m)^{2q} < \infty$$

whenever $\sigma q > 1$, by proposition 3.2 with exponent $2q$.

For necessity when $h \geq 3$, fix a prime p_0 and let $m_r = p_0^{h-1}r$ as $r \neq p_0$ varies over primes. The prime r has exponent one and is not saturated, so $J_h(m_r) = \{p_0\}$. The angle factor in equation (34) is a fixed positive constant and $c_{m_r} \asymp r^{-\sigma}$. The prime sum $\sum_r c_{m_r}^q$ diverges when $\sigma q \leq 1$.

That witness is ill typed for $h = 2$, because exponent one is already saturated. Instead fix p_0 and take $m_r = p_0 r$. Now $J_2(m_r) = \{p_0, r\}$, while

$$1 - (1 - p_0^{-\sigma})(1 - r^{-\sigma}) \geq p_0^{-\sigma}.$$

Again $c_{m_r} \asymp r^{-\sigma}$, proving divergence at and below the same wall.

For M , equation (35) makes the angular factor a fixed positive constant throughout the bounded domain. Thus $c_m \asymp m^{-\sigma}$, and the h -free sum $\sum_m c_m^q$ converges exactly for $\sigma q > 1$. The boundedness condition $\sigma > 1/h$ remains mandatory. The factor two from the two singular values per nonzero block does not affect membership. $\square$

The $h = 2$ case also admits a closed Hilbert–Schmidt identity. For a squarefree m , define

$$\Lambda_m = \prod_{p|m} (p^\sigma - 1)^{-1}, \quad \Delta_m = \prod_{p|m} (1 - p^{-\sigma}).$$

Then $\rho_S(m)^2 = \Lambda_m$ and the two commutator singular values are $\Lambda_m \sqrt{1 - \Delta_m}$.

Proposition 7.3 ($h = 2$ Euler control). *If $\sigma > 1/2$, then*

$$\|[S_{2,s}^*, S_{2,s}]\|_2^2 = 2 \left\{ \prod_p \left[1 + (p^\sigma - 1)^{-2} \right] - \prod_p \left[1 + \frac{p^{-2\sigma}}{1 - p^{-\sigma}} \right] \right\}. \quad (36)$$

Both products converge separately.

Proof. Summing the two squared singular values gives $2 \sum_m \Lambda_m^2 (1 - \Delta_m)$. Squarefree multiplicativity yields the first Euler product from $\sum_m \Lambda_m^2$. The local nontrivial factor in $\Lambda_m^2 \Delta_m$ is

$$(p^\sigma - 1)^{-2} (1 - p^{-\sigma}) = \frac{p^{-2\sigma}}{1 - p^{-\sigma}},$$

which gives the second product. Each nonconstant local factor is $O(p^{-2\sigma})$, so both products converge for $\sigma > 1/2$. At equality, theorem 7.2 already gives divergence; no subtraction of divergent Euler products is defined. $\square$

No ideal endpoint in this section is inferred from a finite block. The strict failure comes from an explicit infinite positive subfamily.

8 Independent recomputation, controls, and limitations

The formulas were recomputed along two deliberately disjoint routes. One route enumerated the raw maps, constructed finite fiber matrices, and computed spectra, singular values, powers, Riesz idempotents, and self-commutators directly. The other route began from prime-exponent states and independently derived Euler factors, strict convergence walls, the Tauberian strip and residue, the primordial asymptotics, and the commutator families. A third checker audited the infinite proof certificates, while a finite comparator inspected only the common projected fields. The finite and analytic routes agreed on their declared overlap, and registered hostile mutations were rejected. Exact case counts, input seals, and finite optimizer rows are reproducibility metadata in appendix D, not premises of theorem 1.1.

This separation matters. A growing finite compression cannot establish boundedness at an infinite fiber, a strict Schatten endpoint, a Tauberian continuation, or uniform boundedness of all Riesz projections. Conversely, the analytic proof does not certify that a software implementation parses the two maps correctly. The two forms of evidence answer different questions.

Free-UFD negative control. Let $\mathfrak{M}$ be the free commutative monoid generated by atoms a_j , and assign $N(a_j) = p_j$, the j th rational prime. Define exponent saturation, reduction modulo h , and the weights through this norm. The relabeling $a_j \leftrightarrow p_j$ reproduces every exponent combinatoric and Euler product used above, even though the atoms carry no addition and no semantics as rational primes. Thus the paired theorem detects normed unique factorization and its two retractions; by itself it does not establish rational-prime selectivity. This control receives no contribution credit.

Scope. The results concern the frozen maps equation (1), the coefficient $n^{-s/2}$, and the counting measure on $\mathbb{N}$. We do not claim a theorem for arbitrary weighted composition operators, changed base measures, other coefficient weights, or non-free factorization systems. Equal regularized determinants are used only to expose cyclic blindness to block geometry. They are not interpreted as a self-adjoint spectral determinant or as a Hilbert–Pólya construction. The equality $C_{h,1} = D_{h,1} = 1$ likewise gives no universal ordering away from the crossover.

Conclusion. The two retractions preserve exactly the same h -free eigenlines but store discarded prime exponents in incompatible fibers. That difference shifts the existence wall from 0 to $1/h$, the saturated similarity wall to 1, and the commutator wall to $1/q$, while leaving the common cyclic ledger unchanged wherever it is legal. The primordial and Weyl laws make

the gap quantitative: isospectrality here is not an approximation to similarity, but a precise separation between arithmetic fixed points and the geometry of their fibers.

A The block graph transform

We record the elementary similarity lemma with explicit constants. Let K be a Hilbert space and

$$T = \begin{pmatrix} \lambda & \varphi \\ 0 & 0 \end{pmatrix} \quad \text{on } \mathbb{C} \oplus K, \quad \lambda \neq 0,$$

where $\varphi : K \rightarrow \mathbb{C}$. Its spectral idempotent at λ is

$$P = \lambda^{-1}T = \begin{pmatrix} 1 & g \\ 0 & 0 \end{pmatrix}, \quad g = \lambda^{-1}\varphi.$$

The row-operator norm gives

$$\|P\| = (1 + \|g\|^2)^{1/2}. \quad (37)$$

Define

$$Y = \begin{pmatrix} 1 & -g \\ 0 & I \end{pmatrix}, \quad Y^{-1} = \begin{pmatrix} 1 & g \\ 0 & I \end{pmatrix}.$$

Then

$$Y^{-1}TY = \begin{pmatrix} \lambda & 0 \\ 0 & 0 \end{pmatrix}, \quad \|Y\|, \|Y^{-1}\| \leq 1 + \|g\|. \quad (38)$$

Thus a uniform bound on the block idempotents is precisely what is needed for the direct sum of the graph transforms to be bounded and invertible.

Conversely, if $T = XNX^{-1}$ and N is compact normal, the Riesz projections Q_m of N at distinct nonzero eigenvalues are orthogonal. The corresponding projections of T are XQ_mX^{-1} , so

$$\sup_m \|XQ_mX^{-1}\| \leq \|X\|\|X^{-1}\|.$$

This proves the necessity and sufficiency used in lemma 4.1 without appealing to a finite-dimensional condition number that may depend on the block.

B Legality of the common regularized determinant

For an integer $r \geq 1$ and $T \in \mathcal{S}_r$, the order- r regularized determinant is standard [Sim05]. In terms of the algebraic nonzero eigenvalues, counted with multiplicity,

$$\det_r(I - zT) = \prod_{\lambda \in \text{spec}(T) \setminus \{0\}} \left[(1 - z\lambda) \exp \left(\sum_{j=1}^{r-1} \frac{(z\lambda)^j}{j} \right) \right]. \quad (39)$$

The empty sum at $r = 1$ gives the Fredholm determinant. Membership in $\mathcal{S}_r$ guarantees convergence and an entire function of z .

For the operators in this paper, proposition 3.2 shows that both $S_{h,s}$ and $M_{h,s}$ lie in $\mathcal{S}_r$ exactly on the common legal domain

$$\sigma > 1/h, \quad r\sigma > 2. \quad (40)$$

Their nonzero eigenvalues are the same simple sequence $\lambda_m = m^{-s/2}$, $m \in \mathcal{F}_h$. Substitution in equation (39) therefore proves

$$\det_r(I - zS_{h,s}) = \det_r(I - zM_{h,s}) \quad (41)$$

on equation (40). When $r = 1$, this condition reads $\sigma > 2$. No product is asserted for either unbounded algebraic map, and determinant equality is not used to infer bounded similarity.

C Tauberian boundary details

We make explicit the analytic inputs behind proposition 6.2. The local factor in equation (25) is entire in z for each fixed p , because the bases are positive and powers use the real logarithm. On a compact set $K \subset \{\Re z > \theta_{h,\sigma}\}$, choose $\varepsilon > 0$ so that

$$h\Re z \geq 1 + \varepsilon, \quad (h-1)\Re z + \sigma \geq 1 + \varepsilon \quad (z \in K).$$

The expansion equation (28) is uniform on K , and the sum of its two majorants is bounded by a constant times $\sum_p p^{-1-\varepsilon}$. Hence the Euler product for $G_{h,\sigma}$ converges normally on K . This proves local uniform convergence and holomorphy, not merely pointwise convergence of formal factors.

At $z = 1$, all local factors are positive and their deviations from one are summable. Thus $G_{h,\sigma}(1) = C_{h,\sigma}$ is finite and strictly positive. Since

$$F_{h,\sigma}(z) = \zeta(z)G_{h,\sigma}(z),$$

$F_{h,\sigma}$ has a simple pole at one with residue $C_{h,\sigma}$ and no other singularity on $\Re z = 1$. Subtracting the polar part gives the boundary regularity required by the Wiener–Ikehara theorem [Kor04].

To connect the series with a monotone function, attach a unit atom to every generalized weight $w_{h,\sigma}(m)$. Only finitely many local weights are below one: for a nonsaturated exponent the weight is p^e , while the saturated weight is

$$p^{h-1}(1-p^{-\sigma})^{1/\sigma} \rightarrow \infty.$$

Because the exponents in $m \in \mathcal{F}_h$ are bounded, removing the finitely many global weights below one changes the count by $O(1)$. The remaining Mellin–Stieltjes transform is exactly $F_{h,\sigma}$ up to an entire finite sum. Wiener–Ikehara therefore yields $A_S(x) \sim C_{h,\sigma}x$.

Finally, if a decreasing positive sequence has counting function $N(t) \sim Ct^{-\alpha}$ as $t \downarrow 0$, elementary monotone inversion gives its n th term as $(C/n)^{1/\alpha}(1+o(1))$. Here $\alpha = 2/\sigma$, which produces equation (29). No numerical coefficient fit is involved.

D Canonical recomputation ledger

This appendix binds the implementation summary used during writing. It is reproducibility metadata, not an additional proof of the infinite theorem. The extraction script verifies every source hash before reading a field and then emits a compact JSON projection and the table below.

D.1 Sealed source artifacts

The frozen experiment contract has SHA-256 `6ff3776a29b1211762b929782b556d0cae71a60ec97b102863059fc5bf302fbc`; the declared fifteen-case infinite set has SHA-256 `6401b141f7b46b0f7275ec124ec571542655b9874cfa9aa5c7123108577e8a84`.

Table 2: SHA-256 seals of the eight canonical result artifacts.

Artifact	SHA-256
SHA256SUMS.txt	2fae66ff866b63e7119fce7b86c928f589570572728cae942d758f4e599ad734
comparator_x.json	6a8404c802342e9ea37fc311ecb23492f2503fa7137258a46b416aafaaee12c5
evaluation_report.json	4c5efa633213cd6f056b550c562dbf9929b3a7145aae33149b482ecd3fec0b5b
evaluator_a.json	ba3f374f1e65e3598c7d4e769144514e911f5be268ae36afc90000df5a5154da
evaluator_b.json	ac8226e8d9a726ebf78e753d66b19e200ba382a5ecdc926ff79a262c7c81a675
integrity_audit.json	fef14966637e160367f545e7c6ee9f53399c6f3de3f6a01b74614ac3bff94c9b
mutation_outcomes.json	8042263d0ddd43b3b2c8c27737c10a053b422e1dfbf9667f292a4b5bba4f147b
proof_auditor_p.json	1a62f35af5b7147599d23139231f17443c14970612ac115387a484b84b60ce4d

D.2 Exact evaluation envelope

The direct map-to-matrix route emitted 21 finite records and no infinite records. The independent exponent-to-Euler route emitted the same 21 finite records and 15 infinite certificates. The proof auditor closed all 15 declared infinite cases with no finding. The finite comparator covered seven case identifiers and found zero exact mismatches and zero certified- interval mismatches. The hostile suite made 168 physical calls covering 75 registered rejection codes and left zero survivor. The exact eight-path output transaction passed its containment and manifest checks, preserved identity under a forced late failure, and made zero physical replacement on an identical second invocation.

The infinite cases separately cover the two existence walls, the power Schatten wall, trace and determinant legality, both similarity conditions, the three primordial regimes, the Tauberian pole and residue, the two Weyl constants and crossover, both commutator endpoint witnesses, the $h = 2$ Euler identity, and the free-UFD negative control.

D.3 Finite primordial checks

Table 3: Exact finite optimizer rows generated from the sealed projection. The supercritical finite row has several ties; the canonical maximizer and the primordial label are both 900.

σ	cutoff x	maximizer	primordial label	ties
2/3	100	36	36	36
1	1000	900	900	900
4/3	10000	900	900	6300, 900, 9900

These three rows test the finite optimizer implementation. They do not establish theorem 5.2, which follows from proposition 5.1 and classical prime asymptotics.

References

- [AM23] A. V. Abanin and R. S. Mannanikov. Weighted composition operators on quasi-Banach weighted sequence spaces. *Vladikavkazskii Matematicheskii Zhurnal*, 25(4):5–19, 2023.
- [Car90] James W. Carlson. The spectra and commutants of some weighted composition operators. *Transactions of the American Mathematical Society*, 317(2):631–654, 1990.
- [dWvdW99] B. M. M. de Weger and C. E. van de Woestijne. On the power-free parts of consecutive integers. *Acta Arithmetica*, 90(4):387–395, 1999.

- [Kor04] Jacob Korevaar. *Tauberian Theory: A Century of Developments*, volume 329 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, 2004.
- [LK15] Doan Minh Luan and Le Hai Khoi. Weighted composition operators on weighted sequence spaces. In *Function Spaces in Analysis*, volume 645 of *Contemporary Mathematics*, pages 199–215. American Mathematical Society, Providence, RI, 2015.
- [MV06] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory*, volume 97 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2006.
- [Sim05] Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2 edition, 2005.